

Embedded Systems Development

Task 5.2C Change the Light Intensity Using GUI

GUIs can be used for more complex tasks such as taking user input and performing a certain task on an embedded system according to the user input. This task combines this capabilities with PWM concepts to improve the features of the user interface.

Hardware Required

- Raspberry Pi with OS installed and remote connections established
- Keyboard, mouse and monitor (HDMI) to plug into the Raspberry Pi.
- LEDs
- Resistors
- Breadboard and jumper wires

Software Required

- Raspberry Pi with OS installed

Pre-requisites

- Complete tasks 5.1P

Embedded Systems Development

Task Objective

Problem statement:

Linda has another request for you. Now that she can Switch ON and OFF the lights using the GUI, she has asked if she can change the intensity levels of the light in her living room.

Solution:

Extend the GUI you developed in Task 5.1 to include a slider switch for the living room, which will allow you to change the light intensity level. Use the learning you gained in PWM and the experience gained in task 5.1P to complete this task. You can use the same programming language and GUI package as 5.1P.

Steps

1. Build a basic breadboard interface with the Raspberry Pi that allows you to turn on or off an LED. You will be using the same set up you used in Task 5.1P. However, be mindful of what PINs support PWM.
2. Build the GUI as described in the task objective.
3. Test the system.
4. **For SIT730 Students**
For two or more lights, add a timer function to start the change of light intensity.

Task Submission Details

- Create a video that shows the outcome of the task. Upload it to YouTube or Deakin Panopto and Include the link here. The video should not exceed 5 minutes and should have the presenter in the view. Please ensure the video contains the following:
 - Overview of how PWM works and how this is used in the task
 - Hardware setup
 - Software code (please do not explain the code line by line. Rather, focus on how different areas of the code or different functions achieve the overall function.)
 - Working system
- Create folder Task5.2GUI in your SIT210/730 GitHub repository. Upload your code to GitHub and include the link to your repository here.